

End-to-End Multimodal Pavement Assessment System

Physics-Informed IRI Sensing, Vision-IMU Pothole Detection & GIS Mapping

Nishkarsh Singhal | B.Tech, Electronics & Communication Engineering

Faculty of Technology, University of Delhi, New Delhi, India | nishkarsh.5108@gmail.com | +91-9205841118

Prepared For:
Dr. Pradeep Kumar
Chief Scientist & Head
Pavement Evaluation Div.
CSIR-CRRI, New Delhi

1. 1. Executive Summary: A Unified Participatory Road Infrastructure Ecosystem

City-scale pavement preservation requires monitoring both **continuous surface roughness (International Roughness Index, IRI)** and **discrete structural failures (potholes/anomalies)**. While Class 1 Network Survey Vehicles (NSVs) provide high-precision reference measurements, high capital costs ($> \$50,000$ USD per vehicle) restrict frequent surveys to arterial corridors. To bridge this gap, we have developed and deployed a comprehensive **Multimodal Road Assessment System** turning passenger vehicles into crowdsourced edge probes. Our platform unifies three core subsystems:

- Continuous IRI Regression Engine:** A speed-invariant deep learning architecture trained via physics simulation to estimate standardized quarter-car IRI from smartphone inertial sensors across heterogeneous vehicle classes.
- Multimodal Vision-IMU Pothole Detector:** An event-level anomaly classification framework fusing dashcam video frames with suspension kinematic impulse signatures via Physics-Informed Cross-Attention.
- Real-Time Android Edge & Cloud GIS Platform:** A deployed mobile application and Geohash-partitioned WebSocket backend enabling live municipal road-condition heatmaps and roughness-aware route planning.

2. 2. Subsystem I: Sim-to-Real Continuous Road Roughness (IRI) Estimation

Empirical smartphone ML models typically overfit to the host vehicle's suspension transfer function ($\mathcal{T}_{\text{veh}}$) and suffer from velocity-squared acceleration distortion ($a_z \propto v^2$). We establish a deterministic Sim-to-Real (Sim2Real) pipeline using the **BeamNG.tech** soft-body physics laboratory across 3 distinct vehicle classes (Roamer SUV, Sunburst Sedan, Vivace Hatchback):

- Curvature-Adaptive Mesh Tessellation Correction:** Solves artificial 3D polygon-facet slope jitter (± 0.3 m/m) on curved grades/ramps by decoupling macro road bending (κ) from true vertical roughness, enabling exact World Bank Golden Car quarter-car IRI calculation (80 km/h reference standard).
- Speed-Invariant 1D-CNN Architecture:** Employs Inverse Quadratic Kinetic Normalization ($22.22/v_{\text{safe}}$)², 2-D inverse-frequency joint histogram sampling, Asymmetric Huber Loss (4 : 1 under-prediction penalty), and Post-Training Isotonic Calibration (PICA).

3. 3. Subsystem II: Multimodal Pothole Detection & Live GIS Edge Deployment

- Physics-Informed Vision-IMU Fusion:** To classify discrete structural failures, we extract a Teager-Kaiser Energy Operator (TKEO) signature isolating damped suspension impact impulses. A **Physics-Informed Cross-Attention (PICA)** layer fuses TKEO features with synchronized dashcam frames conditioned on kinetic energy (E_k) and speed, utilizing Hunter focal loss to suppress hard-braking false positives.
- Live Edge & Route-Planning Platform:** Deployed as an Android edge application backed by a Node.js cloud engine with sub-second Geohash WebSockets, calculating dynamic **Ride Comfort Scores** and computing roughness-aware navigation paths across Delhi NCR expressways (e.g., KMP corridor).
- Multi-Vehicle IRI Performance:** On $N = 10,000$ multi-vehicle test windows, our continuous IRI engine achieves **MAE = 1.642 m/km** (Pearson $r = 0.702$, $R^2 = 0.493$), outperforming CatBoost by 19.1% and Bi-LSTM by 30.9%, with **77.9% AASHTO R 56 compliance**.

Table 1: Continuous IRI Model vs. ML Baselines

Model Architecture	MAE (↓)	RMSE (↓)	R ² (↑)
Ridge Regression ($\alpha = 1.0$)	2.103	2.653	0.110
Random Forest Regressor	2.053	2.581	0.158
CatBoost Ensemble	2.030	2.541	0.185
Standard 1D-CNN (Unscaled)	2.341	2.890	-0.057
Bidirectional LSTM	2.378	2.915	-0.075
Proposed 1D-CNN + PICA	1.642	1.998	0.493

Request for Co-Registered Field Validation Collaboration with CSIR-CRRI

While our complete multimodal edge application and pothole detection subsystems are deployed and tested on public crowdsourced datasets (693 km PVS dataset) and Delhi NCR highways, **rigorous empirical validation of our continuous IRI engine against Class 1 laser profilometer ground truth on Indian roadways is the critical milestone for national infrastructure adoption**. We respectfully seek a brief field collaboration under either modality:

- Option A (Co-Registered NSV Ride-Along):** Accompanying a scheduled CRRI Network Survey Vehicle (NSV) run on a road stretch in Delhi NCR (2–5 km length), recording dashboard smartphone IMU/GPS telemetry to validate model predictions against reference NSV IRI chainage.
- Option B (Reference Chainage Access):** Accessing recent 100-meter segment IRI reference data for a known road stretch near Delhi/Gurugram, allowing us to independently traverse the corridor and report co-registered validation accuracy.

All research publications and institutional acknowledgments will prominently recognize CSIR-CRRI's leadership in advancing scalable, low-cost pavement infrastructure monitoring in India.